Solving degree vs chain depth k: six-point Riemann–Roch decompositions over a 2-adic chain factor base

2026-10-15 (UTC) · study `research/rr_2adic_chain_dreg_20261015/` · experiment log (observations; no hypothesis status changed). Labels: **measured**, **derived**, **literature**, **inference**. Full log: `REPORT.md`.

Question

P-224 has $p-1 = 2^{96}(2^{128}-1)$, P-192 has $p+1 = 2^{64}(2^{128}-1)$. A factor base S with $|S| = 2^k$ x-values has a membership test that is a chain of k squarings. A six-point relation is $f = A(x) + yC(x)$, $\deg A \le 3$, C monic quadratic, $f(-T) = 0$ (5 unknowns), with all six roots of $q = \frac{EC^2 - A^2}{x - x_T}$ in S. An $N^{1/3}$ ECDLP needs $|S| \approx N^{1/6}$ ($k \approx 37$ for P-224) and each system decided in $\approx N^{1/6}/720 \approx 2^{27.8}$ operations. We measure how the Gröbner solving degree grows with k. **Literature:** Petit–Kosters–Messeng, PKC 2016 (doi 10.1007/978-3-662-49387-8_1) is the direct precedent.

Setup

- **Arms:** `mult`, $p = 268496897$ ($2^{12} \mid p-1$), $S = \{X : X^{2^k} = c\}$ (P-224 shape); `dickson`, $p = 268562431$ ($2^{12} \mid p+1$), $S = \{X : D_{2^k}(X) = c\}$ with chain $x \mapsto x^2 - 2$ (P-192 shape, no auxiliary curve).
- **Encodings: direct**, 5 unknowns, equations are the coefficients of $\text{chain}^{k(X)} \bmod q - c$ (input degree 6 at $k = 3$, 22 at $k = 4$); **circuit**, $R_{i+1} = R_i^2 - Q_i q$ with remainder, quotient and q-coefficient variables, all equations of degree ≤ 2, $n = 11k - 19$ variables, $n + 1$ equations.
- **Instances: planted** (positive control; witness checked against every equation) and **random** T (expected relations $\ll 1$).
- **Solver:** msolve 0.10.1, F4, grevlex, 1 thread; Xeon 2.1 GHz, 15 GiB. Metric: max F4 step degree (the solving degree when complete; a lower bound when stopped).
- **Phases:** 1 (05:18Z, $k = 3$–4, 3 seeds, 600 s, 3 jobs); 2 (06:35Z, circuit $k = 4$, 1 job, 2400 s); 3 (08:29Z, circuit $k = 5$–7, mult planted seed 1, 1 job, 2400 s). Code: `build_systems.py` unchanged since commit 6dee0662f2; driver fd0db54170.

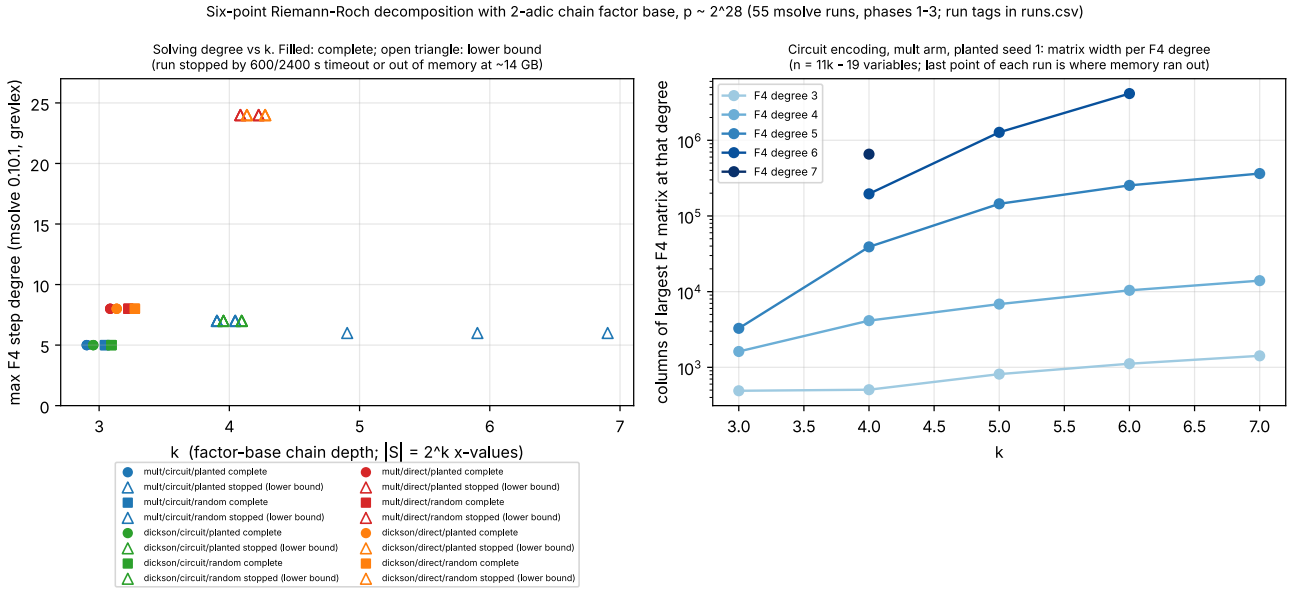


Figure 1: Left: max F4 degree vs k for all 55 runs (filled = complete, open triangle = lower bound from timeout or OOM). Right: columns of the largest F4 matrix per degree, circuit/mult/planted seed 1 (k=3 phase 1, k=4 phase 2, k=5–7 phase 3). Run tags in `runs.csv`, `matrix_by_degree.csv`.

Results (measured)

k	encoding	n	runs	stop	max F4 deg	largest matrix
3	circuit	14	12	complete, ≤ 0.4 s	**5**	3748 × 3275
3	direct	5	12	complete, ≈ 0.2 s	**8**	—
4	direct	5	12	timeout 600 s	≥ 24	5652 × 49785
4	circuit	25	12 (ph. 1)	OOM, 3 jobs	≥ 7	165551 × 195969
4	circuit	25	4 (ph. 2)	OOM, 1 job, 13.6 GB	≥ 7	647230 × 655761
5	circuit	36	1	OOM, 947 s	≥ 6	851697 × 1274254
6	circuit	47	1	OOM, 636 s	≥ 6	2223917 × 4142298
7	circuit	58	1	OOM, 333 s	≥ 6 (step started)	170233 × 363442

Controls (measured): every complete planted run returned exactly 1 solution; every complete random run returned no solution over the closure (`[-1]`); every planted instance passed the witness assertion. Both arms behaved identically at $k = 3$ and $k = 4$.

Derived

- The circuit encoding lowers the solving degree at $k = 3$ from 8 to 5. At $k = 4$ the direct input degree is 22, matching the earlier toy's "22". The circuit's input degree is 2, but its solving degree is ≥ 7.
- Degree-5 matrix width (columns): 3275, 39040, 144749, 253573, 363442 for $k = 3..7$.
- At P-224 size ($k = 37$, $n = 388$), the dense Macaulay width $\binom{n+D}{D}$ is $2^{36.1}$, $2^{42.2}$, $2^{48.0}$ for $D = 5, 6, 7$. Each exceeds the $\approx 2^{27.8}$ per-system budget.

Inference, limitations, next checks

Inference: no measurement shows the solving degree staying near 2–3, which the $N^{1/6}$-per-relation target needs. The values (complete $D = 5$ at $k = 3$; $D \geq 7$ at $k = 4$) are consistent with growth in k. They do not establish a growth law, because no run with $k \geq 4$ completed. **Limitations:** one solver and one order; a 15 GiB memory cap; only the 2^k part of $p \pm 1$; toy $p \approx 2^{28}$; timeouts and OOM kills are operational facts, not refutations. **Next:** finish $k = 4, 5$ on a host with ≥ 64 GB; hybrid guessing of 1–2 unknowns; encodings with fewer variables (eliminate Q_i); a Petit–Kosters–Messeng Semaev-chain baseline with the same S.